

1. [Theory] Which statement best describes an `if-elif-else` chain?
 - A) Every true condition is executed
 - B) Only the first true branch is executed
 - C) Only the last true branch is executed
 - D) All branches are checked and executed
2. [Theory] What is the main purpose of indentation in Python conditional and loop blocks?
 - A) It improves only readability
 - B) It defines the block of statements controlled by the statement
 - C) It converts code into machine language
 - D) It is optional
3. [Interview] What does `break` do inside nested loops?
 - A) Stops all loops
 - B) Stops the nearest enclosing loop
 - C) Skips only the current statement
 - D) Restarts the outer loop
4. [Interview] What does `continue` do in a loop?
 - A) Terminates the program
 - B) Terminates the loop
 - C) Skips the remaining statements of the current iteration and proceeds to the next iteration
 - D) Repeats the current iteration forever
5. [Theory] What is the difference between `=` and `==` in Python?
 - A) Both perform comparison
 - B) `=` assigns a value, while `==` compares values
 - C) `==` assigns and `=` compares
 - D) Both perform assignment
6. [Interview] Which condition is true for an even positive integer `n` greater than 2?
 - A) $n \% 2 == 1$
 - B) $n \% 2 == 0$ and $n > 2$
 - C) $n // 2 == 0$
 - D) $n \% 2 != 0$
7. [Theory] What values are produced by `range(2, 10, 3)`?
 - A) 2, 3, 4, 5, 6, 7, 8, 9
 - B) 2, 5, 8
 - C) 2, 5, 8, 10
 - D) 3, 6, 9
8. [Interview] Which situation can cause a `while` loop to become infinite?
 - A) Its condition eventually becomes False
 - B) The loop variable is updated toward termination

- C) The condition remains True because the relevant state never changes
- D) The loop contains an `if` statement

9. [Theory] In Python, what is the result of `not (True and False)`?

- A) True
- B) False
- C) None
- D) Error

10. [Interview] What is the advantage of using `elif` instead of several independent `if` statements when only one category should be selected?

- A) It guarantees every condition executes
- B) It prevents later branches from being selected after a previous condition succeeds
- C) It removes indentation
- D) It makes loops unnecessary

11. [Theory] What does `//` perform in Python?

- A) Normal division
- B) Modulo
- C) Floor division
- D) Exponentiation

12. [Interview] If `break` occurs inside an inner loop, what normally happens to the outer loop?

- A) The outer loop is also immediately terminated
- B) The outer loop continues normally
- C) The program always terminates
- D) The outer loop becomes infinite

13. [Theory] Which expression checks whether `n` is divisible by both 3 and 5?

- A) `n % 3 == 0 or n % 5 == 0`
- B) `n / 3 == 0 and n / 5 == 0`
- C) `n % 3 == 0 and n % 5 == 0`
- D) `n // 3 == 0 and n // 5 == 0`

14. [Interview] Why is `if x > 10` different from `if x >= 10`?

- A) The first includes 10 and the second excludes it
- B) The first excludes 10 and the second includes it
- C) They are always identical
- D) Both are syntax errors

15. [Theory] What happens when a `for` loop completes normally without encountering `break` and has an `else` clause?

- A) The `else` clause executes
- B) The `else` clause is skipped
- C) The loop restarts
- D) Python raises an error

16. [Interview] What is a common logical mistake when writing a range check for `10 <= x <= 20`?

- A) Using two comparisons
- B) Accidentally using `or` when both boundaries must be satisfied
- C) Using indentation
- D) Using `elif`

17. [Theory] What does `range(5, 0, -1)` generate?

- A) 5, 4, 3, 2, 1
- B) 5, 4, 3, 2, 1, 0
- C) 0, 1, 2, 3, 4
- D) 4, 3, 2, 1, 0

18. [Interview] In a nested `if`, why can the indentation level change the program's logic?

- A) Indentation only changes comments
- B) It determines which statements belong to which conditional block
- C) It changes variable types
- D) It changes arithmetic precedence

19. [Theory] Which operator has higher logical precedence in Python?

- A) `or` has higher precedence than `and`
- B) `and` has higher precedence than `or`
- C) `not` has lower precedence than `or`
- D) All have identical precedence

20. [Interview] What is the best way to trace a difficult loop during an interview?

- A) Guess the output
- B) Track the loop variable and important variables after every iteration
- C) Ignore conditional branches
- D) Count only the first iteration

21. [Output Logic] What is printed?

```
x = 8
if x > 5:
    if x % 2 == 0:
        print("A")
    else:
        print("B")
else:
    print("C")
```

- A) A
- B) B
- C) C
- D) Nothing

22. [Output Logic] What is printed?

```
n = 5
while n > 1:
```

```
if n % 2 == 0:
    n -= 2
else:
    n -= 1
print(n)
```

- A) 0
- B) 1
- C) 2
- D) 3

23. [Output Logic] What is printed?

```
for i in range(1, 5):
    if i == 3:
        continue
    print(i, end=" ")
```

- A) 1 2 3 4
- B) 1 2 4
- C) 2 4
- D) 1 3

24. [Output Logic] What is printed?

```
x = 12
if x % 2 == 0:
    x //= 2
elif x % 3 == 0:
    x += 3
else:
    x -= 1
print(x)
```

- A) 5
- B) 6
- C) 9
- D) 11

25. [Interview Logic] A program must print `Pass` only when marks are at least 40 AND attendance is at least 75. Which condition is correct?

- A) marks >= 40 or attendance >= 75
- B) marks > 40 and attendance > 75
- C) marks >= 40 and attendance >= 75
- D) marks <= 40 and attendance <= 75